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1 High-Precision Sensor Modbus Protocol

2 Command Instructions:

Data format: 8 data bits, no parity, 1 stop bit; default slave address: 0x50

Serial commands must be completed within 10 s; otherwise, the device automatically locks. To avoid automatic locking, follow the steps below.

1. Enter the unlock command
2. Enter the command to be modified
3. Save command

3 Register Table

ADDR (Hex)	ADDR (Dec)	REGISTER NAME	FUNCTION	SERIAL I/F	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit	Bit
					15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
00	00	SAVE	Save / Restart / Restore Factory Settings	R/W	SAVE[15:0]																
01	01	CALSW	calibration mode	R/W														CALSW[3:0]			
04	04	BAUD	serial baud rate	R/W														BAUD[3:0]			
05	05	AXOFFSET	Acceleration X Offset	R/W	AXOFFSET[15:0]																
06	06	AYOFFSET	Acceleration Y Offset	R/W	AYOFFSET[15:0]																
07	07	AZOFFSET	Acceleration Z Offset	R/W	AZOFFSET[15:0]																
08	08	GXOFFSET	Angular Velocity X Offset	R/W	GXOFFSET[15:0]																

09	09	GYOFFSET	Angular Velocity Y Offset	R/W	GYOFFSET[15:0]
0A	10	GZOFFSET	Angular Velocity Z Offset	R/W	GZOFFSET[15:0]
0B	11	HXOFFSET	Magnetic Field X Offset	R/W	HXOFFSET[15:0]
0C	12	HYOFFSET	Magnetic Field Y Offset	R/W	HYOFFSET[15:0]
0D	13	HZOFFSET	Magnetic Field Z Offset	R/W	HZOFFSET[15:0]
0E	14	WORKMODE 【1】	Z-Axis Operating Mode	R/W	WORKMODE[15:0]
10	16	GYROPTP 【1】	Z-Axis Stationary Peak- to-Peak Value	R/W	GYROPTP[15:0]
11	17	GPTPTIME 【1】	Z-Axis Peak-to- Peak Acquisition Time	R/W	GPTPTIME[15:0]
12	18	GYROBAIS 【1】	Z-Axis Zero-Bias Value	R/W	GYROBAIS[15:0]
13	19	GBAISTIME 【1】	Z-Axis Zero-Bias Acquisition Time	R/W	GBAISTIME[15:0]
14	20	GSTATICTHRE 【1】	Z-Axis Stationary Threshold	R/W	GSTATICTHRE[15:0]
15	21	GSTATICTIME 【1】	Z-Axis Stabilization Time	R/W	GSTATICTIME[15:0]
16	22	PGSCALE 【1】	Z-Axis Calibration Factor P	R/W	PGSCALE[15:0]
18	24	GSCALERANGE 【1】	Z-Axis Calibration Angle	R/W	GSCALERANGE[15:0]

33	51	MS	millisecond	R/W	MS[15:0]
34	52	AX	Acceleration X	R	AX[15:0]
35	53	AY	Acceleration Y	R	AY[15:0]
36	54	AZ	Acceleration Z	R	AZ[15:0]
37	55	GX	Angular Velocity X	R	GX[15:0]
38	56	GY	Angular Velocity Y	R	GY[15:0]
39	57	GZ	Angular Velocity Z	R	GZ[15:0]
3A	58	HX	Magnetic Field X	R	HX[15:0]
3B	59	HY	Magnetic Field Y	R	HY[15:0]
3C	60	HZ	Magnetic Field Z	R	HZ[15:0]
3D	61	LRoll	X-Axis Angle Low Word	R	LRoll[15:0]
3E	62	HRoll	X-Axis Angle High Word	R	HRoll[15:0]
3F	63	LPitch	Y-Axis Angle Low Word	R	LPitch[15:0]
40	64	HPitch	Y-Axis Angle High Word	R	HPitch[15:0]
41	65	LYaw	Z-Axis Angle Low Word	R	LYaw[15:0]
42	66	HYaw	Z-Axis Angle High Word	R	HYaw[15:0]
43	67	TEMP	Module Temperature	R	TEMP[15:0]
44~4F			Reserved		

51	81	q0	Quaternion 0	R		
52	82	q1	Quaternion 1	R		
53	83	q2	Quaternion 2	R		
54	84	q3	Quaternion 3	R		
61	97	GYROCALITHR	Gyroscope Calibration Threshold	R/W	GYROCALITHR[15:0]	
63	99	GYROCALTIME	Gyroscope Calibration Time	R/W	GYROCALTIME[15:0]	
69	105	KEY	unlock	R/W	KEY[15:0]	
6A	105	WERROR	Gyroscope Variation Value	R/W	WERROR[15:0]	
6E	110	WZTIME	Z-Axis Stationary Time Threshold	R/W	WZTIME[15:0]	
6F	111	WZSTATIC	Z-Axis Stationary Threshold	R/W	WZSTATIC[15:0]	
7F	127	NUMBERID1	Device ID 1-2	R	ID2[15:8]	ID1[7:0]
80	128	NUMBERID2	Device ID 3-4	R	ID4[15:8]	ID3[7:0]
81	129	NUMBERID3	Device ID 5-6	R	ID6[15:8]	ID5[7:0]
82	130	NUMBERID4	Device ID 7-8	R	ID8[15:8]	ID7[7:0]
83	131	NUMBERID5	Device ID 9-10	R	ID10[15:8]	ID9[7:0]
84	132	NUMBERID6	Device ID 11-12	R	ID12[15:8]	ID11[7:0]
8B	139	MAGANGLE	Magnetometer Angle	R/W		
95	149	LREFROLL	Roll-Angle Zero-Bias Low Word	R/W	LREFROLL[15:0]	

96	150	HREFROLL	Roll-Angle Zero-Bias High Word	R/W	HREFROLL[15:0]
97	151	LREFPITCH	Pitch-Angle Zero-Bias Low Word	R/W	LREFPITCH[15:0]
98	152	HREFPITCH	Pitch-Angle Zero-Bias High Word	R/W	HREFPITCH[15:0]

Note:

1. Marked registers apply only to HWTX073.

4 Read Register Format

- Data are transmitted in hexadecimal format, not as ASCII characters.
- Each register address, the number of registers to read, and the read data are represented by two bytes. The high and low bytes of the register address are represented by ADDR_H and ADDR_L; the high and low bytes of the number of registers to read are represented by LEN_H and LEN_L; and the high and low bytes of the read data are represented by DATA_{1H} and DATA_{1L}.
- The last two bytes of a read command are the standard CRC checksum. A CRC calculator can be used to calculate them; [CRC Online Calculator](#)

Command Request

Modbus Address	Function Code	Register High Byte	Register Low Byte	Read Length High Byte	Read Length Low Byte	CRC High Byte	CRC Low Byte
ID	0x03 (Read)	ADDR _H [15:8]	ADDR _L [7:0]	LEN _H [15:8]	LEN _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data Response

Modbus Address	Function Code	Read Length (8 Bits)	Data High 8 Bits	Data Low 8 Bits	Data High 8 Bits	Data Low 8 Bits	Data High 8 Bits	Data Low 8 Bits	CRC High Byte	CRC Low Byte
ID	0x03 (Read)	LENL[7:0]	DATA1H[15:8]	DATA1L[7:0]			DATAnH	DATAnL	CRCH[15:8]	CRCL[7:0]

Note: All examples below assume the Modbus address is 0x50 (default). If the Modbus address is changed, the address and CRC bytes in the commands must be changed accordingly.

4.1 AX~AZ (Acceleration)

Register name: AX~AZ

Register address: 52~54 (0x34~0x36)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X = AX[15:0]/32768*16g (g is gravitational acceleration; use 9.8 m/s ²)
15:0	AY[15:0]	Acceleration Y = AY[15:0]/32768*16g (g is gravitational acceleration; use 9.8 m/s ²)
15:0	AZ[15:0]	Acceleration Z = AZ[15:0]/32768*16g (g is gravitational acceleration; use 9.8 m/s ²)

Example:

Send: 50 03 00 34 00 03 49 84 (read 3-axis acceleration)

Return: 50 03 06 AXH AXL AYH AYL AZH AZL CRCH CRCL

$AX[15:0] = ((\text{short})AXH \ll 8) | AXL;$

$AY[15:0] = ((\text{short})AYH \ll 8) | AYL;$

$AZ[15:0] = ((\text{short})AZH \ll 8) | AZL;$

4.2 GX~GZ (Angular Velocity)

Register name: GX~GZ

Register address: 55~57 (0x37~0x39)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	GX[15:0]	Angular velocity X = $GX[15:0]/32768*2000^\circ/s$
15:0	GY[15:0]	Angular velocity Y = $GY[15:0]/32768*2000^\circ/s$
15:0	GZ[15:0]	Angular velocity Z = $GZ[15:0]/32768*2000^\circ/s$

Example:

Send: 50 03 00 37 00 03 B9 84 (read 3-axis angular velocity)

Return: 50 03 06 GXH GXL GYH GYL GZH GZL CRCH CRCL

```
GX[15:0]=((short)GXH <<8) | GXL;
```

```
GY[15:0]=((short)GYH <<8) | GYL;
```

```
GZ[15:0]=((short)GZH <<8) | GZL;
```

4.3 HX~HZ (Magnetic Field)

Note: The magnetic-field value calculated from the raw data is in milligauss, which differs from the unit shown in the PC software. To obtain the same unit as the PC software, use the conversion method below.

1. [Magnetic-Field Raw Data Unit Conversion Formula Link](#)

Register name: HX~HZ

Register address: 58~60 (0x3A~0x3C)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	HX[15:0]	Magnetic field X = HX[15:0] (unit: LSB)
15:0	HY[15:0]	Magnetic field Y = HY[15:0] (unit: LSB)
15:0	HZ[15:0]	Magnetic field Z = HZ[15:0] (unit: LSB)

Example:

Send: 50 03 00 3A 00 03 28 47 (read 3-axis magnetic field)

Return: 50 03 06 HXH HXL HYH HYL HZH HZL CRCH CRCL

```
HX[15:0]=((short)HXH <<8) | HXL;
```

```
HY[15:0]=((short)HYH <<8) | HYL;
```

```
HZ[15:0]=((short)HZH <<8) | HZL;
```

4.4 Roll~Yaw (Angles)

[Data Parsing Example Program](#)

Register name: Roll~Yaw

Register address: 61~66 (0x3D~0x42)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	LRoll[15:0]	Roll Angle X Low Word
15:0	HRoll[15:0]	Roll Angle X High Word; Roll Angle X = (((int)HRoll<<16) LRoll)/1000°
15:0	LPitch[15:0]	Pitch Angle Y Low Word
15:0	HPitch[15:0]	Pitch Angle Y High Word; Pitch Angle X = (((int)HPitch<<16) LPitch)/1000°
15:0	LYaw[15:0]	Heading Angle Z Low Word
15:0	HYaw[15:0]	Heading Angle Z High Word; Heading Angle Z = (((int)HYaw<<16) LYaw)/1000°

Example:

Send: 50 03 00 3D 00 06 59 85 (read three-axis angles)

Return: 50 03 0C LRollH LRollL HRollH HRollL LPitchH LPitchL HPitchH HPitchL
LYawH LYawL HYawH HYawL CRCH CRCL

Roll angle (X-axis):

$\text{Roll} = ((\text{HRollH} \ll 24) | (\text{HRollL} \ll 16) | (\text{LRollH} \ll 8) | \text{LRollL}) / 1000^\circ$

Pitch angle (Y-axis):

$\text{Pitch} = ((\text{HPitchH} \ll 24) | (\text{HPitchL} \ll 16) | (\text{LPitchH} \ll 8) | \text{LPitchL}) / 1000^\circ$

Yaw angle (Z-axis): $\text{Yaw} = ((\text{HYawH} \ll 24) | (\text{HYawL} \ll 16) | (\text{LYawH} \ll 8) | \text{LYawL}) / 1000^\circ$

[High-Precision Modbus Protocol Conversion.docx](#)

4.5 TEMP (Temperature)

Register name: TEMP

Register address: 67 (0x43)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature = TEMP[15:0]/100 °C

Example:

Send: 50 03 00 43 00 01 78 5F (read chip temperature)

Return: 50 03 02 TEMPH TEMPL CRCH CRCL

TEMP[15:0]=((short)TEMPH <<8) | TEMPL ;

4.6 q0~q3 (Quaternions)

Register name: q0~q3

Register address: 81~84 (0x51~0x54)

Access: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	q0[15:0]	Quaternion 0 = q0[15:0]/32768
15:0	q1[15:0]	Quaternion 1 = q1[15:0]/32768
15:0	q2[15:0]	Quaternion 2 = q2[15:0]/32768
15:0	q3[15:0]	Quaternion 3 = q3[15:0]/32768

Example:

Send: 50 03 00 51 00 04 18 59 (read quaternion)

Return: 50 03 08 q0H q0L q1H q1L q2H q2L q3H q3L CRCH CRCL

q0[15:0]=((short)q0H <<8) | q0L ;

q1[15:0]=((short)q1H <<8) | q1L ;

q2[15:0]=((short)q2H <<8) | q2L ;

$q3[15:0] = ((\text{short})q3H \ll 8) | q3L ;$

4.7 VERSION (Version Number)

Register name: VERSION

Register address: 46 (0x2E)

Access: R

Default value: None

Bit	NAME	FUNCTION
15:0	VERSION[15:0]	The version number varies by product.

Example:

Read command: 50 03 00 2E 00 02 A9 83

Response data: 50 03 04 81 02 93 9B 1E 51

Response format: 50 03 04 VLH VLL VHH VHL + CRC checksum

Version number $VERSION = (VHH \ll 24) | (VHL \ll 16) | (VLH \ll 8) | VLL$

$VERSION = (0x93 \ll 24) | (0x9B \ll 16) | (0x81 \ll 8) | 0x02$

1001 0011 1001 1011 1000 0001 0000 0010

001 0011 1001 1011 10: 20078

00 0001: 1

0000 0010: 2

Final version: 20078.1.2

5 Write Register Format

- Data are transmitted in hexadecimal format, not as ASCII characters.
- Each register address and the data to be written are represented by two bytes. The high and low bytes of the register address are represented by ADDR_H and ADDR_L, and the high and low bytes of the data to be written are represented by DATA_H and DATA_L.

Command Request

Modbus Address	Function Code	Register High Byte	Register Low Byte	Data High 8 Bits	Data Low 8 Bits	CRC High Byte	CRC Low Byte
ID	0x06 (Write)	ADDR _H [15:8]	ADDR _L [7:0]	DATA _H [15:8]	DATA _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data Response

Modbus Address	Function Code	Register High Byte	Register Low Byte	Data High 8 Bits	Data Low 8 Bits	CRC High Byte	CRC Low Byte
ID	0x06 (Write)	ADDR _H [15:8]	ADDR _L [7:0]	DATA _H [15:8]	DATA _L [7:0]	CRCH[15:8]	CRCL[7:0]

Note: All examples below assume the Modbus address is 0x50 (default). If the Modbus address is changed, the address and CRC bytes in the commands must be changed accordingly.

5.1 KEY (Unlock)

Register name: KEY

Register address: 105 (0x69)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	KEY[15:0]	Unlock register: This register must be set before performing a write operation.

Example:

Send: 50 06 00 69 B5 88 22 A1 (unlock)

Return: 50 06 00 69 B5 88 22 A1

Unlock by writing 0xB588 to this register (all other values are invalid).

5.2 Accelerometer Calibration

Note: During accelerometer calibration, the module must be placed face up. Calibrating it face down may cause abnormal acceleration data and consequently abnormal angle data.

Command procedure:

1. Unlock: 50 06 00 69 B5 88 22 A1

2. Delay 100 ms:

3. Calibrate: 50 06 00 01 00 01 14 4B

4. Delay 2 s:

5. Save: 50 06 00 00 00 00 84 4B

5.3 Angle Reference

The angle-reference function uses the sensor's current physical position as the reference and zeroes the X- and Y-axis angles, providing a relative zeroing operation.

Command procedure:

1. Unlock: 50 06 00 69 B5 88 22 A1

2. Delay 100 ms:

3. Calibrate: 50 06 00 01 00 08 D4 4D

4. Delay 1 s:

5. Save: 50 06 00 00 00 00 84 4B

5.4 Z-Axis Zeroing

Note: Z-axis zeroing requires the 6-axis algorithm. The algorithm can be switched in the PC software configuration interface. For 9-axis devices, the 9-axis algorithm provides an absolute angle and cannot be zeroed.

Command procedure:

1. Unlock: 50 06 00 69 B5 88 22 A1

2. Delay 100 ms

3. Calibrate: 50 06 00 01 00 04 D4 48

4. Delay 1 s:

5. Save: 50 06 00 00 00 00 84 4B

5.5 Change Baud Rate

Command procedure:

1. Unlock: 50 06 00 69 B5 88 22 A1

2. Delay 100 ms

3. Calibrate: 50 06 00 04 00 06 45 88 (change the baud rate to 115200)

3.1 Switch to the modified baud rate, then send the unlock and save commands.

3.2 Unlock: 50 06 00 69 B5 88 22 A1

4. Save: 50 06 00 00 00 00 84 4B

5.6 Automatic Gyroscope Calibration

For automatic gyroscope calibration, 00 = enabled (default) and 01 = disabled.

1. Unlock: 50 06 00 69 B5 88 22 A1

2. Delay 1 s

3. Calibrate: 50 06 00 61 00 01 14 55

4. Save: 50 06 00 00 00 00 84 4B

5.7 SAVE (Save / Restart / Restore Factory Settings)

Register name: SAVE

Register address: 0 (0x00)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Restart: 0x00FF Restore factory settings: 0x0001

Example:

Send: 50 06 00 00 00 FF C4 0B (restart)

Return: 50 06 00 00 00 FF C4 0B

5.8 CALSW (Calibration Mode)

Register name: CALSW

Register address: 1 (0x01)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	CAL[3:0]	Set calibration mode:

		0000(0x00): normal operating mode 0001(0x01): automatic accelerometer calibration 0011(0x03): height zeroing 0100(0x04): heading-angle zeroing 0111 (0x07): Magnetic-field calibration (spherical fitting method) 1000(0x08): set angle reference 1001 (0x09): Magnetic-field calibration (dual-plane mode)
Example: Send: 50 06 00 01 00 04 D4 48 (zero the heading angle) Return: 50 06 00 01 00 04 D4 48		

5.9 BAUD (Serial Baud Rate)

Register name: BAUD		
Register address: 4 (0x04)		
Access: R/W		
Default value: 0x0002		
Bit	NAME	FUNCTION
15:4		

3:0	BAUD[3:0]	Set serial baud rate: 0001 (0x01): 4800bps 0010 (0x02): 9600bps 0011 (0x03): 19200bps 0100 (0x04): 38400bps 0101 (0x05): 57600bps 0110 (0x06): 115200bps 0111 (0x07): 230400bps 1000 (0x08): 460800 bps (supported only by WT931/JY931/HWT606/HWT906) 1001 (0x09): 921600 bps (supported only by WT931/JY931/HWT606/HWT906)
Example: Send: 50 06 00 04 00 06 45 88 (set the serial baud rate to 115200) Return: 50 06 00 04 00 06 45 88		

5.10 AXOFFSET~HZOFFSET (Offset Settings)

Register name: AXOFFSET~HZOFFSET Register address: 5~13 (0x05~0x0D) Access: R/W
--

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AXOFFSET[15:0]	Acceleration X-axis offset; actual acceleration offset = AXOFFSET[15:0]/10000 (g)
15:0	AYOFFSET[15:0]	Acceleration Y-axis offset; actual acceleration offset = AYOFFSET[15:0]/10000 (g)
15:0	AZOFFSET[15:0]	Acceleration Z-axis offset; actual acceleration offset = AZOFFSET[15:0]/10000 (g)
15:0	GXOFFSET[15:0]	Angular-velocity X-axis offset; actual angular-velocity offset = GXOFFSET[15:0]/10000 (°/s)
15:0	GYOFFSET[15:0]	Angular-velocity Y-axis offset; actual angular-velocity offset = GYOFFSET[15:0]/10000 (°/s)
15:0	GZOFFSET[15:0]	Angular-velocity Z-axis offset; actual angular-velocity offset = GZOFFSET[15:0]/10000 (°/s)
15:0	HXOFFSET[15:0]	Magnetic Field X-Axis Offset
15:0	HYOFFSET[15:0]	Magnetic Field Y-Axis Offset
15:0	HZOFFSET[15:0]	Magnetic Field Z-Axis Offset

Example:

Send: 50 06 00 05 03 E8 94 F4 (set the acceleration X-axis offset to 0.1g), 0x03E8=1000, 1000/10000=0.1 (g)

Return: 50 06 00 05 03 E8 94 F4

5.11 WORKMODE (Z-Axis Operating Mode)

Register name: WORKMODE

Register address: 14 (0x0E)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	WORKMODE[3:0]	Set Z-axis operating mode: 0000(0x00): Normal data mode 0001(0x01): Peak-to-peak calculation mode 0010(0x02): Zero-bias calculation mode 0011(0x03): Scale-factor calculation mode

Example:

Send: 50 06 00 0E 00 01 24 48 (automatic zero-bias acquisition)

Return: 50 06 00 0E 00 01 24 48

5.12 GYROPTP (Z-Axis Stationary Peak-to-Peak Value)

Register name: GYROPTP

Register address: 16 (0x10)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	GYROPTP[15:0]	Parameter used for zero-bias calculation. It is acquired automatically by the sensor and does not need to be set.

Write 0x01 to the "WORKMODE" register to enter "Peak-to-Peak Calculation Mode". In this mode, the sensor automatically calculates the maximum and minimum Z-axis angular velocities during the time set by "GPTPTIME" and records them in "GYROPTP". This data is used for zero-bias calculation and filtering in "Zero-Bias Calculation Mode".

Z-axis stationary peak-to-peak value = GYROPTP/1000 (°/s)

Example:

Send: 50 03 00 10 00 01 88 4E (read Z-axis stationary peak-to-peak value)

Return: 50 03 02 00 64 44 63

0x0064=100; Z-axis stationary peak-to-peak value = 100/1000=0.1 (°/s)

5.13 GPTPTIME (Z-Axis Peak-to-Peak Acquisition Time)

Register name: GPTPTIME

Register address: 17 (0x11)

Access: R/W

Default value: 0x000A

Bit	NAME	FUNCTION
15:0	GPTPTIME[15:0]	Peak-to-peak calculation time; default: 10 s.

Example:

Send: 50 06 00 11 00 0A 54 49 (set Z-axis peak-to-peak acquisition time to 10 s)

Return: 50 06 00 11 00 0A 54 49

After entering "Peak-to-Peak Calculation Mode", the difference between the maximum and minimum Z-axis angular velocities during the "GPTPTIME" interval is obtained and stored in "GYROPTP". This data is used for zero-bias calculation and filtering in "Zero-Bias Calculation Mode".

5.14 GYROBAIS (Z-Axis Zero-Bias Value)

Register name: GYROBAIS

Register address: 18 (0x12)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	GYROBAIS[15:0]	Horizontal gyroscope zero-bias value; it can be obtained through "Zero-Bias Calculation Mode".

The Z-axis horizontal gyroscope has a certain zero-bias value when stationary. This value can be used to eliminate angular velocity while stationary. The zero-bias value can be calculated automatically through automatic zero-bias acquisition. In "Zero-Bias Calculation Mode", it is calculated automatically based on the "GYROPTP" peak-to-peak value and the "GBAISTIME" zero-bias acquisition time.

Z-axis zero-bias value = GYROBAIS/1000 (°/s)

Example:

Send: 50 03 00 12 00 01 88 4E (read Z-axis stationary peak-to-peak value)

Return: 50 03 02 00 64 B1 0F

0x0064=100; Z-axis stationary peak-to-peak value = 100/1000=0.1 (°/s)

5.15 GBAISTIME (Z-Axis Zero-Bias Acquisition Time)

Register name: GBAISTIME		
Register address: 19 (0x13)		
Access: R/W		
Default value: 0x000A		
Bit	NAME	FUNCTION

15:0	GBAISTIME[15:0]	Time required to calculate the horizontal gyroscope zero-bias value.
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Example:

Send: 50 06 00 13 00 0A 2F5 89 (set Z-axis zero-bias acquisition time to 10 s)

Return: 50 06 00 13 00 0A 2F5 89

Time required to acquire the Z-axis zero bias. The zero-bias value is acquired according to this duration.

5.16 GSTATICTHRE (Z-Axis Stationary Threshold)

Register name: GSTATICTHRE		
Register address: 20 (0x14)		
Access: R/W		
Default value: 0x0032		
Bit	NAME	FUNCTION
15:0	GSTATICTHRE[15:0]	The larger GSTATICTHRE[15:0] is, the weaker the vibration resistance and the smaller the error. The smaller GSTATICTHRE[15:0] is, the stronger the vibration resistance and the greater the error. Default value: 50

The Z-axis horizontal gyroscope exhibits slight data jitter when stationary.

This parameter can filter out such small fluctuations. When angular velocity is below the value set by "GSTATICTHRE" for the duration set by "GSTATICTIME", the sensor is considered stationary and Z-axis angular velocity is set to zero. In applications where vibration causes Z-axis accumulation, this parameter may be increased appropriately. For very slow uniform rotation, it may be decreased appropriately.

Z-axis zero-bias value = GSTATICTHRE/1000 (°/s)

Example:

Send: 50 06 00 14 00 64 C5 A4 ((set Z-axis stationary threshold to 0.1g),
0x0064=100, 100/1000=0.1(°))

Return: 50 06 00 14 00 64 C5 A4

5.17 GSTATICTIME (Z-Axis Stabilization Time)

Register name: GSTATICTIME

Register address: 21 (0x15)

Access: R/W

Default value: 0x0064

Bit	NAME	FUNCTION
15:0	GSTATICTIME[15:0]]	Z-axis stationary detection time

Z-axis stationary detection time threshold. When angular velocity is below the value set by "GSTATICTHRE" for the duration set by "GSTATICTIME", the sensor is considered stationary. If a shorter stabilization time is required, this parameter may be reduced appropriately. Reducing it can shorten the

stabilization time, but may also increase error.

Z-axis stabilization time = GSTATICTIME/1000 (s)

Example:

Send: 50 06 00 15 00 32 14 5A ((set Z-axis stabilization time to 0.05 s),
0x0032=50, 50/10000=0.05(°))

Return: 50 06 00 15 00 32 14 5A

5.18 PGSCALE (Z-Axis Calibration Factor P)

Register name: PGSCALE

Register address: 22 (0x16)

Access: R/W

Default value: 0x2710

Bit	NAME	FUNCTION
15:0	PGSCALE[15:0]	Range: 0~20000 This parameter is written at the factory using a high-precision turntable. Do not modify it.

The Z-axis gyroscope has measurement error. At the factory, this error is measured using a high-precision turntable and the resulting parameter is written into the sensor. Do not modify this parameter unless necessary. It can be calculated automatically in "Scale-Factor Calculation Mode". After entering this mode, rotate the sensor through the angle set by "GSCALERANGE" to calculate the calibration factor.

Z-axis calibration factor P = PGSCALE/10000.0

Example:

Send: 50 03 00 16 00 01 68 4F (read Z-axis calibration factor P)

Return: 50 03 02 27 74 24 48

0x2774 = 10100; Z-axis calibration factor P = 10100/10000=1.01

5.19 GSCALERANGE (Z-Axis Calibration Angle)

Register name: GSCALERANGE

Register address: 24 (0x18)

Access: R/W

Default value: 0x02D0

Bit	NAME	FUNCTION
15:0	GSCALERANGE[15:0]	This parameter specifies the angle used for scale-factor calibration.

Example:

Send: 50 06 00 18 01 68 04 32 ((set Z-axis calibration angle to 360),
0x0168=360)

Return: 50 06 00 18 01 68 04 32

Rotation angle required in Z-axis "Scale-Factor Calculation Mode"; it is generally set to an integer multiple of 360°.

5.20 IICADDR (Device Address)

Register name: IICADDR

Register address: 26 (0x1A)

Access: R/W

Default value: 0x0050

Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address used for I2C and Modbus communication. 0x01~0x7F

Example:

Send: 50 06 00 1A 00 02 24 4D (set the device address to 0x02)

Return: 50 06 00 1A 00 02 24 4D

5.21 MAGRANGX~MAGRANGZ (Magnetic-Field Calibration Range)

Register name: MAGRANGX~MAGRANGZ

Register address: 28~30 (0x1C~0x1E)

Access: R/W

Default value: 0x01F4

Bit	NAME	FUNCTION
15:0	MAGRANGX[15:0]	Magnetic-field calibration X-axis range
15:0	MAGRANGY[15:0]	Magnetic-field calibration Y-axis range
15:0	MAGRANGZ[15:0]	Magnetic-field calibration Z-axis range
Example: Send: 50 06 00 1C 01 F4 45 9A (set the magnetic-field calibration X-axis range to 500) Return: 50 06 00 1C 01 F4 45 9A		

5.22 BANDWIDTH (Bandwidth)

Register name: BANDWIDTH

Register address: 31 (0x1F)

Access: R/W

Default value: 0x0004

Bit	NAME	FUNCTION
15:4		
3:0	BANDWIDTH[3:0]	Set bandwidth: 0000 (0x00): 256Hz 0001 (0x01): 188Hz

		0010 (0x02) : 98Hz
		0011 (0x03) : 42Hz
		0100 (0x04) : 20Hz
		0101 (0x05) : 10Hz
		0110 (0x06) : 5Hz
Example: Send: 50 06 00 1F 00 01 74 4D (set the bandwidth to 188 Hz) Return: 50 06 00 1F 00 01 74 4D		

5.23 GYRORANGE (Gyroscope Range)

Register name: GYRORANGE		
Register address: 32 (0x20)		
Access: R/W		
Default value: 0x0003		
Bit	NAME	FUNCTION
15:4		
3:0	GYRORANGE[3:0]	Set gyroscope range: 0011 (0x03) : 2000°/s Default: 2000°/s; fixed and not configurable.

Example:

Send: 50 06 00 20 00 03 C5 80 (set the gyroscope range to 2000°/s)

Return: 50 06 00 20 00 03 C5 80

5.24 ACCRANGE (Accelerometer Range)

Register name: ACCRANGE

Register address: 33 (0x21)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	ACCRANGE[3:0]	Set accelerometer range: 0000 (0x00): $\pm 2g$ 0011 (0x03): $\pm 16g$ This parameter cannot be set. The product automatically adapts the acceleration range internally and switches to 16g when acceleration exceeds 2g.

Example:

Send: 50 06 00 21 00 03 94 40 (set the accelerometer range to 16g)

Return: 50 06 00 21 00 03 94 40

5.25 ORIENT (Installation Orientation)

Register name: ORIENT

Register address: 35 (0x23)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:1		
0	ORIENT	Set installation orientation: 0(0x00): horizontal installation 1 (0x01): Vertical installation (the Y-axis arrow of the coordinate system must point upward)

Example:

Send: 50 06 00 23 00 01 B4 41 (set vertical installation)

Return: 50 06 00 23 00 01 B4 41

5.26 AXIS6 (Algorithm)

Register name: AXIS6

Register address: 36 (0x24)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:1		
0	AXIS6	Set algorithm: 0 (0x00): 9-axis algorithm (heading calculated from magnetic field; absolute heading angle) 1 (0x01): 6-axis algorithm (heading calculated by integration; relative heading angle)

Example:

Send: 50 06 00 24 00 01 05 80 (set 6-axis algorithm mode)

Return: 50 06 00 24 00 01 05 80

5.27 FILTK (K-Value Filter)

Register name: FILTK

Register address: 37 (0x25)

Access: R/W

Default value: 0x001E

Bit	NAME	FUNCTION
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		Range: 1~10000; default: 30. Modification is not recommended. If angle performance no longer meets requirements after a change, restore the value to 30.
15:0	FILTK[15:0]	The smaller FILTK[15:0] is, the stronger the vibration resistance and the weaker the real-time response. The larger FILTK[15:0] is, the weaker the vibration resistance and the stronger the real-time response.
Example: Send: 50 06 00 25 00 1E 15 88 (set the K-value filter to 30) Return: 50 06 00 25 00 1E 15 88		

5.28 ACCFILT (Acceleration Filter)

Register name: ACCFILT

Register address: 42 (0x2A)

Access: R/W

Default value: 0x01F4

Bit	NAME	FUNCTION
15:0	ACCFILT[15:0]	Range: 1~10000; default: 500. Modification is not recommended. If angle performance no longer meets requirements after a change, restore the value to 500. The smaller ACCFILT[15:0] is, the stronger the vibration resistance and the slower the real-time response.

		The larger ACCFILT[15:0] is, the weaker the vibration resistance and the faster the real-time response. This parameter is empirical and should be tuned for the actual operating environment. In a tractor environment, ACCFILT[15:0] can be adjusted to 100 because tractor vibration is severe and greater vibration resistance is required.
Example: Send: 50 06 00 2A 01 F4 A5 94 (set the acceleration filter to 500) Return: 50 06 00 2A 01 F4 A5 94		

5.29 YYMM~MS (On-Chip Time)

Register name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	month
7:0	YYMM[7:0]	year
15:8	DDHH[15:8]	hour

7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	second
7:0	MMSS[7:0]	minute
15:0	MS[15:0]	millisecond

Example:

Send: 50 06 00 30 03 16 05 7A (set year/month to 22-03)

Return: 50 06 00 30 03 16 05 7A

Send: 50 06 00 31 09 0C D3 D1 (set day/hour to 12-09)

Return: 50 06 00 31 09 0C D3 D1

Send: 50 06 00 32 3A 1E B7 2C (set minute/second to 30:58)

Return: 50 06 00 32 3A 1E B7 2C

Send: 50 06 00 33 01 F4 74 53 (set milliseconds to 500)

Return: 50 06 00 33 01 F4 74 53

5.30 GYROCALITHR (Gyroscope Stationary Threshold)

Register name: GYROCALITHR

Register address: 97 (0x61)

Access: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	GYROCALITHR[15:0]	Set gyroscope stationary threshold: Gyroscope stationary threshold = GYROCALITHR[15:0]/1000 (°/s)

Example:

Send: 50 06 00 61 00 32 54 40 (set the gyroscope stationary threshold to 0.05°/s)

Return: 50 06 00 61 00 32 54 40

When angular-velocity variation is below 0.05°/s for the duration specified by GYROCALTIME, the sensor identifies the device as stationary and automatically zeros angular-velocity values below 0.05°/s.

The gyroscope stationary threshold can be determined by reading the WERROR register. A typical setting is GYROCALITHR = WERROR * 1.2, in °/s.

This register must be used together with the GYROCALTIME register.

5.31 GYROCALTIME (Gyroscope Auto-Calibration Time)

Register name: GYROCALTIME

Register address: 99 (0x63)

Access: R/W

Default value: 0x03E8

Bit	NAME	FUNCTION
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15:0	GYROCALTIME[15:0]	Set gyroscope auto-calibration time:
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Example:

Send: 50 06 00 63 01 F4 74 42 (set gyroscope auto-calibration time to 500 ms, 0x01F4=500 (ms))

Return: 50 06 00 63 01 F4 74 42

When angular-velocity variation is below GYROCALITHR for 500 ms, the sensor identifies the device as stationary and automatically zeros angular-velocity values below 0.05°/s.

This register must be used together with the GYROCALITHR register.

5.32 WERROR (Gyroscope Variation Value)

Register name: WERROR		
Register address: 106 (0x6A)		
Access: R		
Default value: 0x0000		
Bit	NAME	FUNCTION
15:0	WERROR[15:0]	Gyroscope variation value = $WERROR[15:0]/1000*180/3.1415926$ (°/s) When the sensor is stationary, changes in this register can be used to determine the GYROCALITHR setting.

5.33 WZTIME (Continuous Angular-Velocity Stationary Time)

Register name: WZTIME

Register address: 110 (0x6E)

Access: R/W

Default value: 0x01F4

Bit	NAME	FUNCTION
15:0	WZTIME[15:0]	continuous angular-velocity stationary time

Example:

Send: 50 06 00 6E 01 F4 E5 81 (set continuous angular-velocity stationary time to 500 ms, 0x01F4=500 (ms))

Return: 50 06 00 6E 01 F4 E5 81

When angular velocity is below WZSTATIC for 500 ms, angular-velocity output is set to 0 and the Z-axis yaw angle is not integrated.

This register must be used together with the WZSTATIC register.

5.34 WZSTATIC (Angular-Velocity Integration Threshold)

Register name: WZSTATIC

Register address: 111 (0x6F)

Access: R/W

Default value: 0x012C

Bit	NAME	FUNCTION
15:0	WZSTATIC[15:0]	angular-velocity integration threshold=WZSTATIC[15:0]/1000(°/s)
Example: Send: 50 06 00 6F 01 F4 E5 81 (set angular-velocity integration threshold to 0.5°/s, 0x01F4=500, 500/1000=0.5 (°/s)) Return: 50 06 00 6F 01 F4 E5 81 When angular velocity is greater than 0.5°/s, the Z-axis yaw angle starts integrating the acceleration. When angular velocity is below 0.5°/s for the duration set by WZTIME, angular-velocity output is set to 0 and the Z-axis yaw angle is not integrated. This register must be used together with the WZTIME register.		

5.35 NUMBERID1~NUMBERID6 (Device ID)

Register name: NUMBERID1~NUMBERID6		
Register address: 127~132 (0x7F~0x84)		
Access: R		
Default value: None		
Bit	NAME	FUNCTION
15:0	NUMBERID1[15:0]	

15:0	NUMBERID2[15:0]	
15:0	NUMBERID3[15:0]	
15:0	NUMBERID4[15:0]	
15:0	NUMBERID5[15:0]	
15:0	NUMBERID6[15:0]	
Device number: WT4200000001		

5.36 XREFROLL~YREFPITCH (Angle Zero Reference Values)

Register name: LREFROLL~HREFPITCH

Register address: 149~152 (0x95~0x98)

Access: R/W

Default value: 0x00000

Bit	NAME	FUNCTION
15:0	LREFROLL[15:0]	Roll-angle zero-reference value low word
15:0	HREFROLL[15:0]	Roll-angle zero-reference value high word; roll-angle zero reference = (((int)HREFROLL<<16) LREFROLL)/1000(°)
15:0	LREFPITCH[15:0]	Pitch-angle zero-reference value low word

15:0	HREFPITCH[15:0]	Pitch-angle zero-reference value = $(((\text{int})\text{HREFPITCH} \ll 16) \mid \text{LREFPITCH}) / 1000 (^{\circ})$
Example: The current roll angle is 2°. To set the roll-angle zero reference by subtracting 2°, REFROLL[31:0]=2*1000=2000=0x07D0 LREFROLL[15:0]=0x07D0, HREFROLL[15:0]=0; Send: 50 06 00 95 07 D0 97 CB (set "LREFROLL") Return: 50 06 00 95 07 D0 97 CB Send: 50 06 00 96 00 00 64 67 (set "HREFROLL") Return: 50 06 00 96 00 00 64 67		